HASSE–WITT MATRIX VANISHING AND CUBIC SCALAR CANCELLATION ON THE $k = 3$ QUARTIC REDUCTION BRANCH

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with the comparison of non-ordinary and supersingular reduction primes of abelian varieties, we prove that on the index-7 prime-carry branch at $k = 3$ for the plane quartic family, $k = 3$ forces vanishing of the Hasse–Witt trace condition $H_p = 0$ and exact cancellation of the full cubic scalar concomitant whenever $H_p = 0$.

1. Main Proposition

Proposition 1.1. *For primes $p \equiv 6 \pmod 7$ on the $k = 3$ branch of the quartic family, the Cartier–Manin/Hasse–Witt invariant H_p satisfies the exact polynomial factorization in which the cubic scalar concomitant vanishes identically modulo H_p.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000786 in the domain of arithmetic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped arithmetic lemma* deciding the scope: k=3 implies H_p=0 and exact cancellation of the full cubic scalar concomitant modulo H_p.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- all-prime non-ordinary vs. supersingular reduction comparison.

References

[1] Primary problem list and source literature for Result 20000786 (arithmetic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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